Personalized predictive public transport crowding information with automated data sources

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Abstract

The paper proposes a methodology for providing personalized, predictive in-vehicle crowding information to public transport travellers via mobile applications or at-stop displays. Three crowding metrics are considered: (1) the probability of getting a seat on boarding, (2) the expected travel time standing, and (3) the excess perceived travel time compared to uncrowded conditions. The methodology combines prediction models of passenger loads and alighting counts based on lasso regularized regression and multivariate PLS regression, a probabilistic seat allocation model and a bias correction step in order to predict the crowding metrics. Depending on data availability, the prediction method can use a combination of historical passenger counts, real-time vehicle locations and real-time passenger counts. We evaluate the prediction methodology in a real-world case study for a bus line in Stockholm, Sweden. The results indicate that personalized, predictive crowding information that is robust to varying data availability can be provided sufficiently early to be useful to travellers. The methodology is of value for agencies and operators in order to increase the attractiveness and capacity utilization of public transport.

Keywords: public transport, crowding, real-time crowding information, prediction

1 Introduction

Efficient, convenient and reliable public transport is a vital part of a sustainable urban transport system. Crowding inside vehicles has negative effects

on all these aspects. Traveler satisfaction and comfort are known to decrease at higher crowding levels, in particular for passengers who are not able to get a seat [1, 2]. For buses in particular, dwell time impacts of crowding due to longer boarding and alighting times are substantial [3]. There is also a negative feedback loop between crowding and arrival time irregularity, which leads to deteriorating passenger waiting times and experienced crowding [4, 5]. All in all, the negative consequences of crowding inhibit the transition from private to public transport in urban areas.

As the urban populations grow, the need increases to utilize available capacity in the transport system more efficiently. There are multiple supply-side methods for combating in-vehicle crowding, ranging from the strategic planning level (increased service frequency, vehicle capacities, network expansions, etc.) to the real-time control level (holding, stop-skipping, extra departures, etc.) [6, 7, 8]. Measures that target the travel demand, meanwhile, are less developed.

The ongoing digitalization of the public transport system has lead to the emergence of several valuable technologies, including automated vehicle locations (AVL), automated passenger counts (APC) and automated fare collection (AFC) systems [9]. In recent years, many cities have started to utilize AVL data for providing real-time bus arrival time information at stops and in mobile applications [10]. Real-time arrival time information systems have positive effects on traveller satisfaction [11], perceived waiting times [12], safety and security [13], and service disruption impacts [14]. In this context several methods for short-term arrival time prediction have been proposed [15, 16, 17].

Systems providing real-time crowding information (RTCI) for buses are still rare. For urban rail transit systems, meanwhile, RTCI is being introduced by public transport agencies and private mobility information providers [18, 19]. RTCI provision allows travellers to make better informed decisions about whether to board a vehicle or not based on their preferences for crowding, waiting time, walking distance, etc. [20, 21]. RTCI may thus directly increase passenger satisfaction as well as indirectly increase satisfaction and service quality by reducing negative crowding externalities.

1.1 Literature review

Providing timely crowding information to passengers waiting at or arriving to a stop generally requires prediction of crowding conditions based on the most recent information about the system state. The problem of predicting passenger loads on buses downstream of their current locations based on

real-time AVL and AFC data is considered in [22]. The prediction is performed in a sequence of two steps, where the first step involves identifying the historical day most similar to the current day and predicting downstream loads based on observed loads from the historical day. In the second step the passenger loads are updated by combining real-time and historical data in an extended Kalman filter (e.g., [23]). The methodology is evaluated on data from a bus line in Shenzhen, China. In [24] bus load factors (loads relative to seated capacity), discretized into six categories, are predicted based on observed load factors of preceding vehicle departures from the same stop using a Markov chain approach. The methodology is applied to APC data from a bus line in Cracow, Poland.

In the context of trains, a Bayesian approach to predicting the passenger loads in individual train cars downstream of their current locations based on APC data is proposed in [25]. Passenger OD flows are assumed to follow a Poisson prior distribution with parameters estimated from historical data, and the posterior distribution is updated based on current-day observed loads. The approach, which also predicts boarding and alighting counts, is evaluated in a simple synthetic case study. A long short-term memory (LSTM) encoder-predictor neural network [26] for predicting train loads based on temporal features (day-of-week, time-of-day, etc.) and current-day load measurements from previous train departures at the same station is proposed in [27]. The method is applied to a railway line in Paris using boarding and alighting data from on-board radar sensors.

In [28] several regression models (stepwise linear regression, lasso regularized linear regression and boosted regression tree ensembles [29]) combining real-time and historical APC data are used to predict individual train car loads, and evaluates the performance on a metro line in Stockholm, Sweden. The lasso regression model is applied to bus load prediction and extended to incorporate real-time AVL data in [30].

A framework for urban rail real-time crowding prediction and information provision based on AFC data is proposed in [31]. The methodology combines data-driven prediction of origin-destination passenger flows using random forests and boosted regression trees [29], with on-line simulation of supply and demand interactions, to predict crowding and service performance. The framework is applied to parts of the underground network in London. Incoming passenger flows to metro stations, aggregated in 15-min intervals, based on AFC data from preceding time intervals are predicted in [32]. A gradient boosted regression trees model [29] is trained using historical AFC data from Shenzhen, China. Passenger loads and on-board crowding are not considered.

Some papers consider public transport ridership prediction for short-term planning rather than real-time information applications. Several neural network architectures are used in [33] to predict daily ridership on a train line in Taiwan based on temporal features, but do not consider real-time data or load variations along the line. A combination of AFC data, network assignment and elasticity parameters is used in [34] to predict passenger load changes due to strategic decisions such as frequency and route changes. The number of public transport trips to special event locations (e.g., stadiums) is predicted in [35] based on AFC data and local event information obtained from the Internet. The developed model is designed to predict arrivals several days in advance.

Simple prediction schemes based on the crowding of the one or two most recent vehicle runs [36] as well as more complex schemes involving running the simulation model forward to a fixed point solution [37] have been proposed and evaluated using simulation models. Both studies demonstrate that predictive RTCI facilitates a more even crowding distribution among metro runs and reduces passengers' experienced travel time.

Previous studies provide valuable knowledge about the ability to predict passenger loads based on combinations of historical and real-time data and the effectiveness of different prediction techniques. However, several research gaps can be identified. Most importantly, existing studies [22, 24, 25, 27, 28, 30, 31] have focused on predicting passenger loads (in some case also boarding and alighting counts [25, 31]). From a passenger's perspective, however, the load alone does not capture the experienced crowding conditions throughout the trip. In particular, most passengers perceive a distinct difference between travelling seated and standing [38]. Assuming available seats will be filled by standing travellers, the chances of getting a seat depend not only on the load but on the boarding and alighting counts at every stop between the passenger's boarding and alighting stops. In other words, two passengers in the same vehicle may experience the crowding differently depending on where they boarded. There is a need to predict personalized crowding indicators that take into account the passenger's origin and destination stops and the dynamically changing seat availability.

Second, previous papers have generally assumed a homogeneous set of data sources for all vehicles. However, it is of practical importance to handle heterogeneous data availability and facilitate crowding prediction when sensor equipment varies among vehicles. Operators and agencies should also be able to assess prediction performance under different data scenarios and the benefits of investing in additional sensors or communication technology that may be required. In particular for urban bus services, where passen-

ger demand influences travel times and regularity, there is potential of using predictors from real-time AVL data combined with historical load data for crowding prediction as a substitute and complement to real-time load data. So far only [30] has considered real-time AVL predictors and only for passenger load prediction.

1.2 Aim of the paper

This paper makes the following contributions:

- A framework for personalized predictive crowding information provision from user inputs (time, origin stop and destination stop) and automated public transport data sources is proposed. The traveller can request the information for a specific trip, customized to the traveller's crowding preferences, and disseminated back via mobile applications or at-stop displays.

- The paper moves beyond passenger load as crowding indicator and considers three metrics involving seat availability: (1) the probability of getting a seat on boarding, (2) the expected travel time standing, and (3) the excess perceived travel time compared to uncrowded conditions. Metrics (2) and (3) consider an aggregate of the entire passenger journey from origin stop to destination stop. Metric (3) further incorporates the subjective perception of crowding, which can be customized to the individual traveller. The traveller-specific crowding metrics are computed from predicted loads and alighting counts.

- Two methods for predicting both loads and alighting counts along the passenger journey are proposed. First, the paper extends a linear regression model of passenger loads [30] to alighting counts and develops the sets of predictors for both load and alighting count prediction. The method facilitates multiple combinations of data sources: historical load data, real-time AVL data and real-time APC data. A large number of variables are considered out of which a subset are selected through lasso regularization and used in predictions. Second, a multivariate partial least squares (PLS) regression model is introduced.

- The effect on prediction performance of different levels of data availability and prediction horizons is evaluated in a real-world case study of an urban bus line. The spatial patterns of the personalized crowding metrics across origin and destination stops are compared to the passenger loads.

The paper is organized as follows. Section 2 presents the methodology including the proposed personalized crowding metrics, the probabilistic seat allocation model, the proposed prediction method, and the considered crowding predictors. Section 3 introduces the real-world case study of a bus line in Stockholm, Sweden, including the utilized data and the model fitting. Section 4 presents the results of the case study, and Section 5 concludes the paper.

2 Methodology

This section first outlines the proposed process for computing predictive passenger-oriented crowding metrics based on automatic passenger count (APC) and vehicle location (AVL) data. The following subsections describe the set of passenger-oriented crowding metrics and the probabilistic seat allocation model. The section then shows how the crowding metrics for a particular vehicle trip can be computed based on predicted passenger loads and alighting counts. A regularized regression prediction method that limits the risk of overfitting is proposed, and sets of predictors based on historical APC data and real-time AVL and APC data are introduced. The notation used in the paper is listed in Table 1.

2.1 Predictive personalized crowding information framework

The predictive crowding information provision process is triggered when a passenger requests information about the predicted crowding conditions for a particular vehicle trip. Depending on data availability, historical APC data, real-time AVL data and real-time APC data relevant for the vehicle trip are retrieved from database and fed into the prediction module where loads and alighting counts are predicted. Based on these the seat allocation module calculates the probability of sitting or standing on each journey segment from origin to destination, since this cannot be directly observed from the data. The predicted sitting and standing probabilities, predicted loads and user-input time multipliers, representing the perceived cost of crowding, are used to calculate the crowding metrics for the vehicle trip. In the final step, a bias correction module removes systematic bias caused by non-linear dependencies between crowding metrics and loads and alighting counts. The process is illustrated in Figure 1.

Table 1: Notation

i	Vehicle trip index, $i \in I$
k	Stop/segment index, $k = 1, \ldots, K$
o	Traveller's origin stop
d	Traveller's destination stop
s	Prediction source stop
m	Crowding level, $m = 1, \ldots, M$
PSB_{io}	Probability of getting a seat upon boarding
ETTS_{iod}	Expected travel time standing
EPTT_{iod}	Excess perceived travel time
π_{ik}^{inv}	Probability of getting a seat at k if standing
$P_{ik\mid o}^{\text{std}}$	Probability of standing on segment k given boarding at o
τ_{ik}	In-vehicle travel time on segment k
T_{iod}	In-vehicle travel time between stops o and d
h_{ik}	Headway to preceding vehicle
q_{ik}	Passenger load at departure
b_{ik}	Number of boarding passengers
a_{ik}	Number of alighting passengers
$\bar{q}_{ik}^{\text{tod}}$	Time-of-day-specific historical mean load
$\bar{a}_{ik}^{\text{tod}}$	Time-of-day-specific historical mean alighting count
$\bar{q}_{ik}^{\text{wday}}$	Weekday-specific historical mean load
$\bar{a}_{ik}^{\text{wday}}$	Weekday-specific historical mean alighting count
$\bar{q}_{ik}^{\text{mnth}}$	Month-specific historical mean load
$\bar{a}_{ik}^{\text{mnth}}$	Month-specific historical mean alighting count
$\hat{q}_{ik\mid s}$, $\hat{a}_{ik\mid s}$	Predicted bus load, alighting count for stop k from stop s
c^{sit}	Vehicle seat capacity
$\mathcal{I}_{ikm}^{\text{lf}}$	1 if crowding at level m, 0 otherwise
ϕ_m	Lower bound of crowding level m
β_m^{sit}, β_m^{std}	Time multipliers sitting and standing at crowding level m
I^{trn}	Prediction models training trip set
N^{trn}	Number of training set trips
$\mathbf{x}_{isk}$	Load and alighting counts predictor vector
$\mathbf{v}_{sk}$, $\mathbf{w}_{sk}$	Coefficient vectors for load and alighting count models
λ_{sk}, γ_{sk}	Regularization hyperparameters for load and alighting count models
$\mathbf{x}_{ik}^{\text{hist}}$, $\mathbf{x}_{is}^{\text{avl}}$, $\mathbf{x}_{is}^{\text{apc}}$	Historical APC, real-time AVL and real-time APC predictor vectors

2.2 Personalized crowding metrics

A range of crowding metrics have been proposed in the literature, some targeted towards planners and operators and others towards travellers [39]. The focus here is on personalized metrics, i.e., metrics that are relevant for an individual traveller making a journey between a particular origin stop and destination stop. Furthermore, we consider both objective metrics and met-

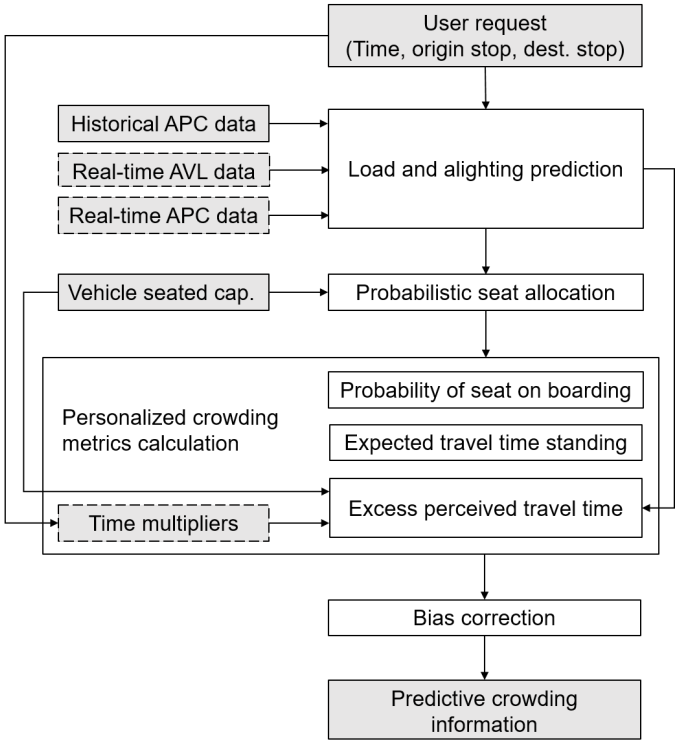


Figure 1: Framework for personalized predictive crowding information provision. Grey boxes indicate data input and output, dashed boxes indicate data that are optional depending on availability.

rics that incorporate subjective valuations of varying crowding conditions. The metrics are also chosen to be as easy as possible to interpret for the traveller.

2.2.1 Probability of seat on boarding

A clear distinction can be made between travelling seated and travelling standing. The value of having a seat varies from being a necessity for some travellers, to a minor comfort factor for others. In particular for the first category, a relevant indicator of in-vehicle crowding conditions is the chance of getting a seat immediately when boarding the vehicle, denoted

$$\mathrm{PSB}_{io} \tag{1}$$

for any trip $i \in I$ and journey origin stop $o \in \{1, \ldots, K-1\}$.

2.2.2 Expected travel time standing

For travellers who prefer sitting over standing but do not strictly require a seat the whole trip, a relevant metric of in-vehicle comfort is the expected travel time standing, i.e., the time until getting a seat or until alighting, whichever occurs first. This crowding metric depends on both the origin and the destination of the traveller. Given the probability of getting a seat on boarding PSB_{io} and the in-vehicle probability π^{inv}_{ik} of getting a seat at stop k, the probability that the probe traveller is standing on line segment $k \in \{o, \ldots, d-1\}$ is equal to the probability that the traveller does not get a seat at stop k or any preceding stop,

$$P^{\mathrm{std}}_{ik|\,o} = (1 - \mathrm{PSB}_{io}) \prod_{k'=o+1}^{k} (1 - \pi^{\mathrm{inv}}_{ik'}). \tag{2}$$

Given the in-vehicle travel time τ_k on segment k, the expected travel time standing is computed as

$$\mathrm{ETTS}_{iod} = \sum_{k=o}^{d-1} \tau_{ik} P^{\mathrm{std}}_{ik|\,o} \tag{3}$$

for any journey between origin stop $o \in \{1, \ldots, K-1\}$ and destination stop $d \in \{o+1, \ldots, K\}$.

2.2.3 Excess perceived travel time

Travellers experience time differently depending on the circumstances under which the time is spent [40, 41]. Several studies have estimated travellers' willingness to pay for shorter travel times under different crowding conditions, typically expressed as time multipliers to in-vehicle time under nominal conditions [38, 42, 43, 44]. We assume here that crowding conditions are discretized into M levels based on the load factor q_{ik}/c^{sit}. The indicator variable $\mathcal{I}^{\mathrm{lf}}_{ikm}$ is 1 if the experienced crowding conditions are at level m and 0 otherwise,

$$\mathcal{I}^{\mathrm{lf}}_{ikm} = \begin{cases} 1 & \phi_m \le q_{ik}/c^{\mathrm{sit}} < \phi_{m+1}, \\ 0 & \text{otherwise.} \end{cases} \tag{4}$$

for a sequence of thresholds ϕ_m such that $\phi_1 = 0$, $\phi_m < \phi_{m+1}$ for every $m \in \{1, \ldots, M\}$. Crowding valuation studies show that the perceived travel time between any two stops depends on whether the traveller is seated or

standing. Let $\beta_m^{\text{sit}} \geq 1$ and $\beta_m^{\text{std}} \geq 1$ denote the time multipliers for in-vehicle time seated and standing, respectively, at crowding level m.

The *excess perceived travel time* is the difference between the expected perceived in-vehicle travel time across travelling standing and seated and the nominal in-vehicle travel time without crowding from origin stop to destination stop. The parameters β_m^{sit} and β_m^{std} could be taken from planning standards, or customized by the individual traveller. The latter approach yields a highly personalized crowding metric, capturing the characteristics of the particular journey as well as the attitudes of the traveller.

Considering that the probability of standing on trip segment k is given in Eq. (2), the excess perceived travel time EPTT_{iod} is obtained as the expectation over both seated and standing states and summation over all trip segments,

$$\text{EPTT}_{iod} = \sum_{k=o}^{d-1} \tau_{ik} \sum_m \mathcal{I}_{ikm}^{\text{lf}} \left(P_{ik|\,o}^{\text{std}} \beta_m^{\text{std}} + (1 - P_{ik|\,o}^{\text{std}}) \beta_m^{\text{sit}} \right) - T_{iod} \qquad (5)$$

2.3 Probabilistic seat allocation

The seated status of the traveller is modelled probabilistically consistent with observed passenger loads and boarding and alighting counts. To harmonize with the crowding prediction framework, the model from [45] is reformulated entirely in terms of passenger loads and alighting counts. The model makes the same seat priority assumptions as [46] and [47]. Thus, sitting passengers are guaranteed a seat until alighting, and standing passengers are given priority over boarding passengers to the seats that become available when sitting passengers alight. These assumptions may not always be accurate in practice; specifically for short trips, some standing travellers may choose to remain standing and leave available seats to boarding passengers if they are only a few stops from their destination stop. In any case, such behaviour implies that the travellers perceive travelling standing similarly to travelling seated. In the absence of research evidence on the topic we opt for the simpler model here.

The number of boarding and alighting passengers, respectively, at stop k are denoted b_{ik} and a_{ik}, and the passenger load on the line segment between stops k and $k+1$ is denoted q_{ik}. The boarding and alighting counts and on-board loads are related as $q_{ik} = q_{i,k-1} + b_{ik} - a_{ik}$. Using this identity and eliminating b_{ik} the seat allocation model can be expressed entirely in terms of loads and alighting counts. It is convenient to define $q_{i,0} = 0$ for any trip i as the load prior to departure from the first stop.

Seat availability at boarding depends on the existing in-vehicle passenger load and the number of boarding passengers. The probability that the traveller receives a seat is 0 if the passenger load before boarding, $q_{i,k-1} - a_{ik}$, exceeds the seat capacity c^{sit}, and is 1 if the passenger load after boarding, q_{ik}, is lower than c^{sit}. Otherwise, the probability of receiving a seat is modelled as the ratio of available seats to the number of boarding passengers,

$$\text{PSB}_{ik} = \begin{cases} 0 & q_{i,k-1} - a_{ik} > c^{\text{sit}} \\ \frac{c^{\text{sit}} - q_{i,k-1} + a_{ik}}{q_{ik} - q_{i,k-1} + a_{ik}} & q_{i,k-1} - a_{ik} \leq c^{\text{sit}},\ q_{ik} > c^{\text{sit}} \\ 1 & q_{ik} \leq c^{\text{sit}}. \end{cases} \tag{6}$$

Inside the vehicle, seats that become available as passengers alight are filled by randomly selected standing passengers. Conditional on that the traveller is standing on segment $k-1$, the probability of getting a seat on segment k is 1 if the passenger load before boardings is lower than the seated capacity. In general, given that a seated passengers alight, the probability of getting a seat is

$$\pi^{\text{inv}}(a, q_{i,k-1}, a_{ik}) = \begin{cases} \frac{a}{q_{i,k-1} - a_{ik} - c^{\text{sit}} + a} & q_{i,k-1} - a_{ik} > c^{\text{sit}} \\ 1 & q_{i,k-1} - a_{ik} \leq c^{\text{sit}}. \end{cases} \tag{7}$$

In the absence of passenger origin-destination information, the number a is not observed. Here, the alighting passengers are randomly selected among all on-board passengers. This assumption implies that the number of seated passengers alighting a is drawn among all seated and standing passengers according to a hypergeometric distribution with population size $q_{i,k-1}$, number of success states c^{sit} and number of draws a_{ik},

$$P(a \mid q_{i,k-1}, a_{ik}) = \frac{\binom{c^{\text{sit}}}{a}\binom{q_{i,k-1} - c^{\text{sit}}}{a_{ik} - a}}{\binom{q_{i,k-1}}{a_{ik}}}, \quad a \in \{0, \ldots, a_{ik}\}. \tag{8}$$

The probability of getting a seat is obtained by summing over all possible a,

$$\pi_{ik}^{\text{inv}} = \begin{cases} \sum_{a=0}^{a_{ik}} P(a \mid q_{i,k-1}, a_{ik}) \cdot \pi^{\text{inv}}(a, q_{i,k-1}, a_{ik}) & q_{i,k-1} - a_{ik} > c^{\text{sit}}, \\ 1 & q_{i,k-1} - a_{ik} \leq c^{\text{sit}}. \end{cases} \tag{9}$$

Figure 2 illustrates the different crowding indicators for a particular passenger trip on bus line 4 in Stockholm, Sweden based on AVL and APC data. The solid black line shows that the probability of getting a seat on boarding is 0. The seat probability then gradually increases as passengers

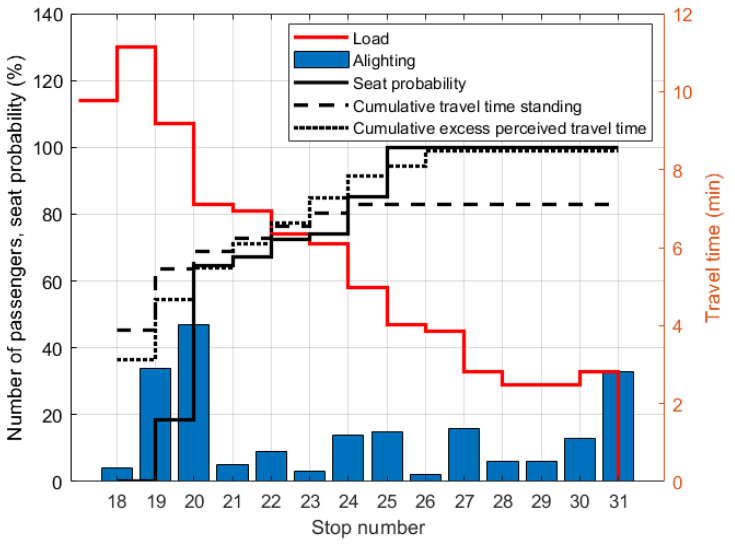


Figure 2: Crowding indicators for a particular trip on bus line 4, Stockholm (26 February 2016, departure 16:48) from origin stop 18 (Västerbroplan) to destination stop 31 (Gullmarsplan).

alight and seats become available. The dashed black line shows the cumulative expected travel time standing, which remains constant from the stop where the passenger is sure to get a seat to the destination stop. The dotted black line shows the cumulative excess perceived travel time based on time multipliers from [38]. This crowding indicator continues to increase even as the passenger is seated, since the perceived in-vehicle time also depends on the load factor.

2.4 Predictive crowding information

Consider a traveller who intends to board the public transport line at origin stop o at time t and travel to destination stop d. We let s denote the source stop, i.e., the most recently visited stop from which real-time data are available for the relevant vehicle at the time $t' < t$ when the traveller requests the crowding information. To provide information to the traveller prior to boarding about expected crowding conditions, the crowding metrics must be predicted based on information available at time t'.

This paper utilizes the fact that all proposed crowding metrics can be computed based on passenger loads and alighting counts between the trav-

eller's origin and destination stops. Specifically, Eq. (6) shows that the probability of seat on boarding crowding metric at origin stop o is a non-linear function of $q_{i,o-1}$, q_{io} and a_{io},

$$\mathrm{PSB}_{io} = \mathrm{PSB}(q_{i,o-1}, q_{io}, a_{io}) \tag{10}$$

Similarly, inserting Eqs. (6) and (9) in (2), and then inserting (2) in (3), shows that the expected travel time standing crowding metric is implicity a non-linear function of loads and alighting counts between stops $o-1$ and $d-1$,

$$\mathrm{ETTS}_{iod} = \mathrm{ETTS}_{iod}\left(q_{i,o-1}, \ldots, q_{i,d-1},\ a_{io}, \ldots, a_{i,d-1}\right), \tag{11}$$

Analogously, the excess perceived travel time crowding metric can be expressed as a non-linear function of the same loads and alighting counts,

$$\mathrm{EPTT}_{iod} = \mathrm{EPTT}_{iod}\left(q_{i,o-1}, \ldots, q_{i,d-1},\ a_{io}, \ldots, a_{i,d-1}\right). \tag{12}$$

The crowding metrics are predicted by substituting predicted passenger loads and alighting counts into the crowding metric definitions. The approach allows multiple crowding metrics to be calculated without the need for multiple predictions. Furthermore, the predicted excess perceived travel time metric can easily be personalized to different traveller's crowding perceptions by combining predicted load factors with customized time multipliers.

Figure 3 illustrates the main concepts of the approach. The predicted load and alighting count on trip i at stop k given information from source stop s are denoted $\hat{q}_{ik|\,s}$ and $\hat{a}_{ik|\,s}$, respectively. The earlier that the traveller requests the crowding information, the longer distance between the source stop s and origin stop o. The predicted probability of seat on boarding is obtained by inserting the predicted loads and alighting counts in Eq. (10),

$$\hat{\mathrm{PSB}}_{io|\,s} = \mathrm{PSB}\left(\hat{q}_{i,o-1|\,s}, \hat{q}_{io|\,s},\ \hat{a}_{io|\,s}\right). \tag{13}$$

In the same way, the predicted travel time standing is

$$\hat{\mathrm{ETTS}}_{iod|\,s} = \mathrm{ETTS}_{iod}\left(\hat{q}_{i,o-1|\,s}, \ldots, \hat{q}_{i,d-1|\,s},\ \hat{a}_{io|\,s}, \ldots, \hat{a}_{i,d-1|\,s}\right), \tag{14}$$

and the predicted excess perceived travel time is

$$\hat{\mathrm{EPTT}}_{iod|\,s} = \mathrm{EPTT}_{iod}\left(\hat{q}_{i,o-1|\,s}, \ldots, \hat{q}_{i,d-1|\,s},\ \hat{a}_{io|\,s}, \ldots, \hat{a}_{i,d-1|\,s}\right). \tag{15}$$

The latter two metrics require predicted segment travel times τ_{ik} between the origin stop and the destination stop. In this study the historical mean travel time on each segment is used to compute the crowding metrics.

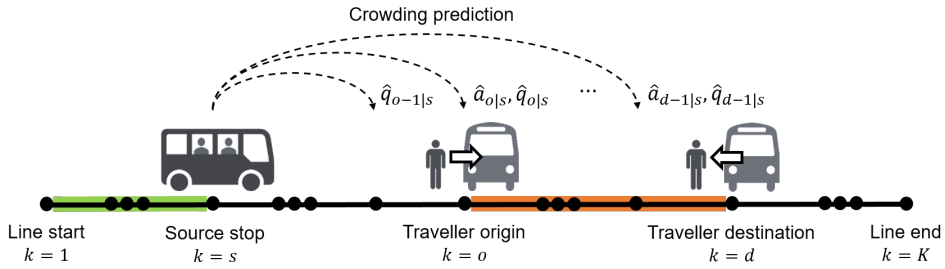


Figure 3: Illustration of main concepts in the prediction framework.

2.5 Prediction method

The machine learning literature contains a large number of prediction methods of varying strengths, complexity and transparency. This section presents two approaches to predicting the passenger loads and alighting counts, the first based on univariate lasso regularized linear regression [29] and the second based on multivariate partial least squares regression [48]. After applying one of the two methods, loads and alighting counts are adjusted iteratively to produce a feasible and consistent load profile between the passenger's origin and destination stops.

2.5.1 Lasso regularized regression

Lasso regularization is a method for avoiding overfitting of model parameters. Similar to ridge regression and support vector machine (SVM) regression, it penalizes large parameter estimates. However, while ridge and SVM regression use a Euclidean (L^2) norm penalty term which generally maintains non-zero estimates, lasso regularization uses a "Manhattan" (L^1) norm penalty term which enforces non-influential predictor coefficients to be zero. In this way, lasso regularization simultaneously performs feature selection and parameter estimation. The feature selection mechanism allows the model training to start from a gross set of features common for all source and target stops and find a subset of relevant predictors customized for each specific source and target stop combination. The linear model formulation allows for transparent and easily interpretable models.

The passenger load $\hat{q}_{ik|s}$ and alighting count $\hat{a}_{ik|s}$ at stop k are both assumed to be linear univariate functions of a common set of predictors $\mathbf{x}_{isk}$ whose values depend on the vehicle trip i and source stop s. The predicted

load is

$$\hat{q}^0_{ik|\,s} = \mathbf{x}_{isk}\mathbf{w}^T_{sk}. \tag{16}$$

The parameter vector $\mathbf{w}_{sk}$, specific to each combination of stop k and source stop s, is estimated so as to minimize the regularized mean squared error,

$$\frac{1}{2}\sum_{i\in I^{\text{trn}}}\left(q_{ik} - \mathbf{x}_{isk}\mathbf{w}^T_{sk}\right)^2 + \gamma_{sk}\left\|\mathbf{w}_{sk}\right\|_1, \tag{17}$$

on the training data set, where $\|\cdot\|_1$ is the "Manhattan" norm. The regularization hyper-parameter $\gamma_{sk} \geq 0$ penalizes large parameter values. Larger γ_{sk} enforce sparser models with higher degrees of freedom, i.e., more parameters equal to zero. The γ_{sk} value is calibrated using cross-validation separately for each (s, k) combination to minimize the cross-validation mean squared error.

Similarly, the predicted alighting count is

$$\hat{a}^0_{ik|\,s} = \mathbf{x}_{isk}\mathbf{v}^T_{sk}. \tag{18}$$

The parameter vector $\mathbf{v}_{sk}$ is estimated on training data to minimize the regularized mean square error,

$$\frac{1}{2}\sum_{i\in I^{\text{trn}}}\left(a_{ik} - \mathbf{x}_{isk}\mathbf{v}^T_{sk}\right)^2 + \gamma_{sk}\left\|\mathbf{v}_{sk}\right\|_1, \tag{19}$$

where $\gamma_{sk} \geq 0$ is the regularization hyper-parameter.

2.5.2 Multivariate partial least squares regression

In partial least squares (PLS) regression, both predictors and response variables are projected to a latent space of dimension lower than the original number of observations and predictors. The method thus combines dimensionality reduction similar to principal component analysis (PCA) with regression, such that the covariance between predictors and response variables is maximized [48]. The dimensionality reduction handles issues of multicollinearity among predictors and can be seen as a form of regularization. Unlike the univariate lasso regression model parameters are estimated jointly for multiple response variables, which allows correlations between loads and alighting counts at different stops to be considered.

For each trip i the passenger loads and alighting counts for all stops downstream of source stop s are collected in a $1 \times 2(K - s)$ multivariate response variable vector,

$$\mathbf{y}_{is} = \left(q_{i,s+1}, \ldots, q_{iK},\ a_{i,s+1}, \ldots, a_{iK}\right). \tag{20}$$

The predictor variables are collected analogously,

$$\mathbf{x}_{is} = (\mathbf{x}_{is,s+1}, \ldots, \mathbf{x}_{isK})\,. \tag{21}$$

All training data are collected in matrices $\mathbf{Y}_s = (\mathbf{y}_{is})_{i \in I^{\text{trn}}}$ and $\mathbf{X}_s = (\mathbf{x}_{is})_{i \in I^{\text{trn}}}$. The PLS model is specified as

$$\mathbf{X}_s = \mathbf{T}_s \mathbf{P}_s^T + \mathbf{E}_s, \tag{22}$$

$$\mathbf{Y}_s = \mathbf{U}_s \mathbf{Q}_s^T + \mathbf{F}_s, \tag{23}$$

where $\mathbf{T}_s$ and $\mathbf{U}_s$ are matrices with p_s columns each containing a score vector for a latent factor, and the columns of $\mathbf{T}_s$ are orthogonal. Further, $\mathbf{P}_s$ and $\mathbf{Q}_s$ are matrices containing corresponding factor loadings, and $\mathbf{E}_s$ and $\mathbf{F}_s$ are matrices of residuals. The value of p_s can be calibrated using cross-validation.

Employing the SIMPLS algorithm proposed by [49], the score matrix $\mathbf{T}_s$ is computed as $\mathbf{T}_s = \mathbf{X}_s \mathbf{R}_s$, where $\mathbf{R}_s$ is a weight matrix in which the p_s columns are computed through a sequence of projections and singular value decompositions of the product $\mathbf{X}_s^T \mathbf{Y}_s$. As shown in [48], the model can be written as a direct linear relation between predictors and responses,

$$\mathbf{Y}_s = \mathbf{X}_s \mathbf{B}_s + \mathbf{F}_s^*, \tag{24}$$

where $\mathbf{B}_s = \mathbf{R}_s \mathbf{T}_s^T \mathbf{Y}_s$. The predicted passenger loads and alighting counts for trip i are

$$\hat{\mathbf{y}}^0_{is|\,s} = \mathbf{x}_{is} \mathbf{B}_s, \tag{25}$$

where $\hat{\mathbf{y}}^0_{is|\,s} = \left(\hat{q}^0_{i,s+1|\,s}, \ldots, \hat{q}^0_{iK|\,s},\ \hat{a}^0_{i,s+1|\,s}, \ldots, \hat{a}^0_{iK|\,s}\right)$.

2.5.3 Iterative adjustment

In the second step, the iterative adjustment process is initialized by rounding the initial passenger load to the nearest non-negative integer,

$$\hat{q}_{i,o-1|\,s} = \max\left\{\text{round}\left(\hat{q}^0_{i,o-1|\,s}\right), 0\right\}. \tag{26}$$

The process then alternatively adjusts alighting counts and loads progressively along the line. Since the number of alighting passengers cannot exceed the load at departure from the preceding stop, the alighting counts are adjusted as

$$\hat{a}_{ik|\,s} = \min\left\{\max\left\{\text{round}\left(\hat{a}^0_{ik|\,s}\right),\ 0\right\},\ \hat{q}_{i,k-1|\,s}\right\}, \quad k \in \{o, \ldots, d-1\}. \tag{27}$$

Since the passenger load at departure cannot be lower than the load before boardings (i.e., the number of boarding passengers must be non-negative), the loads are adjusted as

$$\hat{q}_{ik|\,s} = \max\left\{\text{round}\left(\hat{q}^{0}_{ik|\,s}\right),\ \hat{q}_{i,k-1|\,s} - \hat{a}_{ik|\,s}\right\}, \quad k \in \{o, \ldots, d-1\}. \tag{28}$$

The iterative process is repeated until the last trip segment $d-1$ before the destination stop d.

2.6 Crowding predictors

The proposed prediction method is versatile in terms of the type of variables that are used to predict the passenger loads and alighting counts. We consider three main categories of predictor variables: variables based on (i) historical passenger count data, (ii) real-time vehicle location data, and (iii) real-time passenger count data. The specific variables used in each category are described below. These variables represent a gross list of potential predictors; as explained above the lasso regression maintains only a subset of relevant variables in the final regression model.

2.6.1 Historical APC data

The prediction can utilize temporal patterns in passenger counts at the target stop extracted from historical data. This may be the only available information unless vehicles are equipped with real-time AVL and APC systems. Even with such systems installed, only historical information are available before the vehicles depart from the first stop on the line. Further, historical averages provide a robust baseline from which real-time data can add additional information.

It is well known that passenger counts tend to vary systematically with the time of day, day of week, time of year, etc. Specifically, as predictors for the load at stop k of each vehicle trip i we consider the historical mean loads for departures during the same time-of-day interval $\bar{q}^{\text{tod}}_{ik}$, weekday $\bar{q}^{\text{wday}}_{ik}$ and month $\bar{q}^{\text{mnth}}_{ik}$. We also include an interaction term $\bar{q}^{\text{tod}}_{ik} \cdot \bar{q}^{\text{wday}}_{ik} \cdot \bar{q}^{\text{mnth}}_{ik}$ which captures potential interdependence between the time-of-day, weekday and monthly patterns.

Analogously, the historical mean alighting counts for departures during the same time-of-day interval $\bar{a}^{\text{tod}}_{ik}$, weekday $\bar{a}^{\text{wday}}_{ik}$ and month $\bar{a}^{\text{mnth}}_{ik}$ and an interaction term $\bar{a}^{\text{tod}}_{ik} \cdot \bar{a}^{\text{wday}}_{ik} \cdot \bar{a}^{\text{mnth}}_{ik}$ are considered as predictors for the alighting count. To allow for greater flexibility in the fitting of both load

and alighting count prediction models, all historical predictors are collected in the common 1×8 vector

$$\begin{aligned}\mathbf{x}_{ik}^{\text{hist}} = \Big(&\bar{q}_{ik}^{\text{tod}}, \bar{q}_{ik}^{\text{wday}}, \bar{q}_{ik}^{\text{mnth}},\ \bar{q}_{ik}^{\text{tod}} \cdot \bar{q}_{ik}^{\text{wday}} \cdot \bar{q}_{ik}^{\text{mnth}},\\ &\bar{a}_{ik}^{\text{tod}}, \bar{a}_{ik}^{\text{wday}}, \bar{a}_{ik}^{\text{mnth}},\ \bar{a}_{ik}^{\text{tod}} \cdot \bar{a}_{ik}^{\text{wday}} \cdot \bar{a}_{ik}^{\text{mnth}}\Big).\end{aligned} \tag{29}$$

2.6.2 Real-time AVL data

Vehicle location data are currently more commonly available in real time than passenger counts. Given that the target vehicle has departed from the terminus, AVL data up to the source stop s can be utilized. Although AVL data contain no load data in themselves, they may contain useful information for load prediction, particularly for bus services where stop dwell times are strongly influenced by passenger demand. We consider two types of variables.

First, the more passengers boarding a vehicle, the longer the dwell time at the stop [50]. Thus, the total run time $T_{i,1,s}$ from the start of the line to the source stop is an indicator of the passenger load on the vehicle. Traffic conditions may also influence the run time, which weakens the predictive power of the indicator.

Second, for high-frequency public transport services, passenger arrival times are known to be well approximated by a Poisson process [6]. Under such conditions, the expected number of passengers boarding a vehicle at a given stop is proportional to the headway to the preceding vehicle. Headway is therefore an indicator of passenger load. Here, we include the headway h_{ik} at the K_s most recently visited stops upstream of and including source stop s.

To allow for some non-linear effects we also include the squared version of all variables. Other functional forms than quadratic terms would also be possible to consider (e.g., exponential or logarithmic). For headways, linear and quadratic terms are associated with mean values and variability of crowding, respectively (e.g., [51]). This gives a $1 \times (2K_s + 2)$ vector of predictor variables,

$$\begin{aligned}\mathbf{x}_{is}^{\text{avl}} = (&T_{i,1,s},\ h_{i,s-K_s+1}, \ldots, h_{is},\\ &T_{i,1,s}^2,\ h_{i,s-K_s+1}^2, \ldots, h_{is}^2).\end{aligned} \tag{30}$$

2.6.3 Real-time APC data

With real-time APC data, the load and alighting count are predicted based on the load q_{is}, boarding count b_{is} and alighting count a_{is} at the source

stop. Further, to utilize the fact that passengers' destinations may vary systematically with their origins, we include the boarding counts b_{ik} and alighting counts a_{ik} at the $K_s - 1$ most recently visited stops upstream of s. To allow for non-linear effects we also include the squared version of all variables. In total, we consider the $1 \times (4K_s+2)$ vector of real-time passenger count predictors,

$$\begin{aligned}\mathbf{x}_{is}^{\text{apc}} = (&q_{is},\ a_{i,s-K_s+1}, \ldots, a_{is},\ b_{i,s-K_s+1}, \ldots, b_{is},\\ &q_{is}^2,\ a_{i,s-K_s+1}^2, \ldots, a_{is}^2,\ b_{i,s-K_s+1}^2, \ldots, b_{is}^2)\,.\end{aligned} \tag{31}$$

2.7 Bias correction

The regularized least squares model estimation ensure that predicted loads and alighting counts have low bias. Still, the crowding metrics may be biased since they are non-linear functions of predicted loads and alighting counts (compare Eqs. (13)–(15)). In this paper we consider a simple correction method for the expected travel time standing and the excess perceived travel time by subtracting the mean error on the training set from the baseline prediction for each combination of source, origin and destination stop. Thus, the bias corrected predictions of the two crowding metrics are

$$\hat{\hat{\text{ETTS}}}_{iod|\,s} = \hat{\text{ETTS}}_{iod|\,s} - \delta_{ods}^{\text{ETTS}} \tag{32}$$

$$\hat{\hat{\text{EPTT}}}_{iod|\,s} = \hat{\text{EPTT}}_{iod|\,s} - \delta_{ods}^{\text{EPTT}}, \tag{33}$$

where $\delta_{ods}^{\text{ETTS}}$ and $\delta_{ods}^{\text{EPTT}}$ are the mean errors on the training set,

$$\delta_{ods}^{\text{ETTS}} = \frac{1}{N^{\text{trn}}} \sum_{i \in I^{\text{trn}}} \left(\hat{\text{ETTS}}_{iod|\,s} - \text{ETTS}_{iod} \right) \tag{34}$$

$$\delta_{ods}^{\text{EPTT}} = \frac{1}{N^{\text{trn}}} \sum_{i \in I^{\text{trn}}} \left(\hat{\text{EPTT}}_{iod|\,s} - \text{EPTT}_{iod} \right). \tag{35}$$

Here I^{trn} is the training set of observations and N^{trn} the number of training observations.

3 Case study

This section describes a real-world application of the personalized crowding prediction methodology, including the considered scenarios of data availability, a description of the bus line, an account of the data sources and the model fitting procedure.

3.1 Levels of data availability

Three scenarios of data availability are considered. The most basic scenario is where only historical load data are available. Thus, predictions must be based fully on historical temporal patterns. In this case, the potential predictors $\mathbf{x}_{isk}$ are the 1×9 vector

$$\mathbf{x}_{isk} = \left(1, \mathbf{x}_{ik}^{\text{hist}}\right). \tag{36}$$

In the second scenario, historical load data are complemented with real-time AVL data. In this case, $\mathbf{x}_{isk}$ is the $1 \times (2K_s + 11)$ vector

$$\mathbf{x}_{isk} = \left(1, \mathbf{x}_{ik}^{\text{hist}}, \mathbf{x}_{is}^{\text{avl}}\right). \tag{37}$$

In the third scenario, representing the highest level of data availability, historical load data and real-time AVL data are complemented with real-time APC data. In this case, $\mathbf{x}_{isk}$ is the $1 \times (6K_s + 13)$ vector

$$\mathbf{x}_{isk} = \left(1, \mathbf{x}_{ik}^{\text{hist}}, \mathbf{x}_{is}^{\text{avl}}, \mathbf{x}_{is}^{\text{apc}}\right). \tag{38}$$

3.2 Bus line characteristics

The crowding prediction is applied to the north-to-south direction of high-frequency bus line 4 in Stockholm, Sweden, shown in Figure 4. Line 4 is 12.4 km long and has 31 stops in the north-south direction. The travel time from start to end is typically around 60 minutes. Parts of the route are equipped with dedicated bus lanes and/or transit signal priority. To a large extent, however, the buses run in dense mixed traffic. The line is serviced with 4–6 minutes planned headway during work hours. The operations during this time period are regularity-based as opposed to following a fixed schedule, i.e., each bus driver seeks to maintain equal headways to the preceding and following buses on the same line. According to the public transport planning guidelines for Stockholm [52], the seated capacity of the type of articulated buses used on line 4 is $c^{\text{sit}} = 45$ passengers.

The line has around 30,000 boarding passengers per day and per direction, and it is common particularly during peak hours to not get a seat. Figure 5, top left, shows the distribution of passenger loads and alighting counts along the line for the 30th, 60th and 90th percentile levels across all trips in the data set. There are substantial variations across stops and three peaks dispersed along the line. The other three diagrams illustrate the distributions of the three crowding metrics considered in this paper in terms

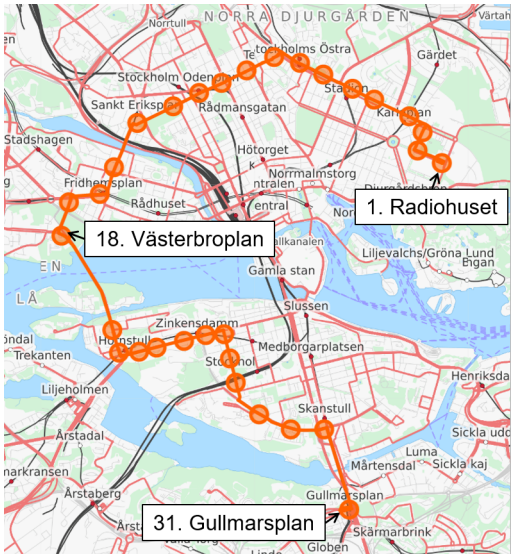


Figure 4: Route of bus line 4. Studied direction is north-south from Radiohuset to Gullmarsplan. Map source: OpenStreetMap.

of the same percentiles. The distributions differ substantially as the probability to get a seat on boarding (top right) reflects local crowding conditions at each stop, the expected travel time standing (bottom left) is affected by crowding conditions in a neighborhood of downstream segments, while the excess perceived travel time (bottom right) reflects cumulative crowding conditions over the entire trip. It can be noted that at most stops a boarding passenger is more likely than not to receive a seat immediately, but the seat probability varies significantly along the line (top right diagram).

3.3 Data and model fitting

The study considers the afternoon peak from 15:00 to 17:30, Mondays through Fridays during 2016. Periods with lower service frequency, including summer, holidays and weekends, are excluded. AVL data are available for all buses and all stops on line 4 during this period. APC data are available only for about 20% of all bus trips. The sensors counting boarding and alighting passengers are installed on a random sample of all buses in Stockholm, and are moved between vehicles at regular intervals. Since the vehicles equipped with APC sensors are selected at random, there is no systematic bias in the selection of days for the evaluation of prediction performance. The data is

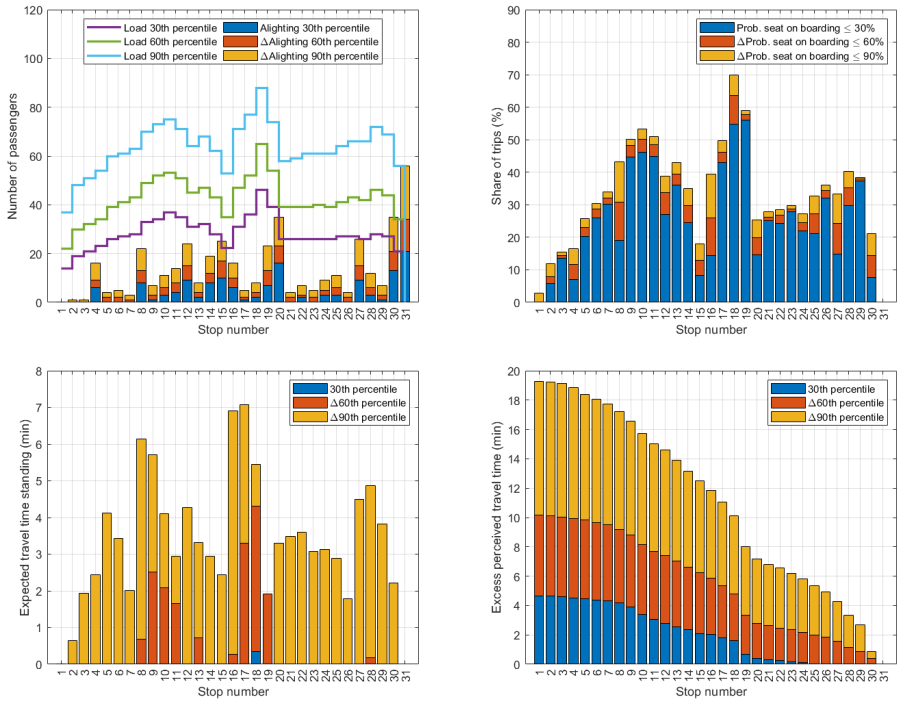


Figure 5: Passenger-oriented crowding metrics, percentiles across all trips in the data set. Top left: Passenger load and alighting count across stops. Top right: Probability of getting a seat on boarding across origin stops. Bottom left: Expected travel time standing across origin stops on trip to end stop Gullmarsplan. Bottom right: Perceived excess travel time across origin stops on trip to end stop Gullmarsplan.

currently collected in batches, but is used here to evaluate the potential of having AVL and APC data available in real-time.

There are in total 5861 unique bus trips across 191 days in the data set. Of these, 1200 trips (20.5%) have recorded APC data. The other runs are excluded from the analysis; however, headways are computed based on the complete set of trips.

Out of the 190 days, 95 days (471 bus runs with recorded load data) are randomly selected as the test set while the remaining 95 days (475 bus runs with load data) are used as training set. Historical mean loads and alighting counts are calculated across all 190 days. The number of stops upstream of

Table 2: Time multiplies for seated and standing in-vehicle time at different crowding levels [38]. In the case study all values have been divided by β_1^{sit}.

Crowding level m	Load factor q/c^{sit} (%)	β_m^{sit}	β_m^{std}
1	0–75	0.86	—
2	75–100	0.95	—
3	100–125	1.05	1.62
4	125–150	1.16	1.79
5	150–175	1.27	1.99
6	175–200	1.40	2.20
7	200–	1.55	2.44

and including the source stop s to include is set to $K_s = \min\{s, 6\}$. For the lasso regression we use 10-fold cross-validation based on the mean squared error to select the regularization coefficient λ for each model. The number of latent dimensions in the PLS regression is set to $p_s = 5$ for the historical load data scenario, $p_s = \min(40, N_s)$ for the historical load and real-time AVL data scenario, and $p_s = \min(60, N_s)$ for the historical load and real-time AVL and APC data scenario, where N_s is the number of predictor variables.

For the excess perceived travel time, the time multipliers used for in-vehicle time at different crowding levels are the multipliers for commuters taken from the meta-study [38]. They are listed in Table 2.

4 Results

This section presents results from the application of the personalized crowding information prediction methodology to bus line 4 in Stockholm. Presented results focus on the lasso regression prediction approach, which is compared to the PLS regression approach in Section 4.6. To gain an understanding of which historical and real-time features are most relevant, we first investigate the selection frequency of different predictors across all source and target stops. The next subsection evaluates the prediction performance of loads and alighting counts across all stops and in detail for a specific stop. The remaining subsections investigate the prediction performance for each of the three personalized crowding metrics. Two main dimensions are studied: the impact of data availability (the three levels introduced in Section 3.1) and prediction horizon (10 and 1 minutes before boarding) on the ability to correctly predict the experienced crowding conditions.

4.1 Predictor relevance analysis

As described in Section 2.6, each lasso regularized regression model selects a subset of relevant predictors from a gross set of potential predictors; non-relevant predictors are excluded by estimating their coefficients to zero. Historical predictors are included in the gross set of every model. In the real-time data scenarios, source-station predictors are also included in every model gross set. Predictors with longer lags are restricted to fewer models; predictors for stop $s-1$ are only included for s greater than 1, and so on. To investigate the relevance of each predictor across all target stops and source stops, we compute the relative frequency with which each predictor is selected from the gross set across all models.

Table 3 shows the relative selection frequency in the load and alighting count models, respectively, for each level of data availability. Values above 0.5 are indicated in bold to highlight the most relevant predictors. The bottom row of the table shows the average number of predictors included in each model relative to the maximum possible number.

Among the historical predictors, the interaction variables are by far the most frequently included. Load-related predictors are sometimes included in models targeting alighting counts and vice versa. It is notable that historical predictors are frequently included even when real-time AVL and APC predictors are available. This suggests taht historical temporal patterns provide an important baseline for predictions.

Among the AVL predictors the source stop headway is the most frequently included predictor for both loads and alighting counts. Headways with longer lags are often included but the frequency tends to decline with the lag. Total run time is included more often than not when real-time APC predictors are not available, but appears to be less important in general than the headway. Squared predictors are less commonly included except, interestingly, for the longest lag headway, which may serve as a collector of circumstances further upstream. Not surprisingly, AVL predictors are less frequently included when APC predictors are available. Thus, AVL serves partially as a proxy for APC.

As expected, the most frequently included APC predictor is the source stop passenger load. Boarding predictors are included somewhat more frequently than alighting predictors. This is intuitive since passengers boarding have a more direct effect on downstream loads and alighting counts. Selection frequencies are quite stable across lags.

Table 3: Relative selection frequency of model predictors. Values above 0.5 are indicated in bold.

Predictor	Hist Target		Hist+AVL Target		Hist+AVL+APC Target	
Hist	$\hat{q}_{ik}$	$\hat{a}_{ik}$	$\hat{q}_{ik}$	$\hat{a}_{ik}$	$\hat{q}_{ik}$	$\hat{a}_{ik}$
$\bar{q}_{ik}^{\text{tod}}$	0.133	0.133	0.193	0.274	0.283	0.161
$\bar{q}_{ik}^{\text{wday}}$	0.333	0.133	**0.524**	0.326	0.352	0.161
$\bar{q}_{ik}^{\text{mnth}}$	0.267	0.200	0.435	0.299	0.212	0.097
$\bar{q}_{ik}^{\text{tod}} \cdot \bar{q}_{ik}^{\text{wday}} \cdot \bar{q}_{ik}^{\text{mnth}}$	**1.00**	0.200	**0.995**	0.338	**0.984**	0.267
$\bar{a}_{ik}^{\text{tod}}$	0.233	0.233	0.322	0.269	0.255	0.195
$\bar{a}_{ik}^{\text{wday}}$	0.367	0.333	**0.613**	0.490	0.379	0.324
$\bar{a}_{ik}^{\text{mnth}}$	0.267	0.233	0.460	0.391	0.161	0.239
$\bar{a}_{ik}^{\text{tod}} \cdot \bar{a}_{ik}^{\text{wday}} \cdot \bar{a}_{ik}^{\text{mnth}}$	0.367	**0.967**	0.345	**1.00**	0.218	**0.991**
AVL						
$T_{i,1,s}$			**0.589**	**0.568**	0.283	0.202
$T_{i,1,s}^2$			0.391	0.347	0.149	0.193
h_{is}			**0.975**	**0.878**	**0.917**	**0.676**
h_{is}^2			0.421	0.267	0.140	0.106
$h_{i,s-1}$			**0.707**	**0.603**	0.419	0.345
$h_{i,s-1}^2$			0.335	0.249	0.148	0.118
$h_{i,s-2}$			**0.545**	0.447	0.233	0.225
$h_{i,s-2}^2$			0.400	0.320	0.246	0.143
$h_{i,s-3}$			0.496	0.453	0.180	0.188
$h_{i,s-3}^2$			0.319	0.282	0.242	0.160
$h_{i,s-4}$			0.443	0.431	0.206	0.197
$h_{i,s-4}^2$			0.440	0.348	0.345	0.212
$h_{i,s-5}$			0.430	0.410	0.363	0.293
$h_{i,s-5}^2$			**0.533**	0.480	0.450	0.280
APC						
q_{is}					**0.752**	**0.632**
q_{is}^2					0.235	0.269
b_{is}					0.455	0.453
b_{is}^2					0.202	0.212
a_{is}					0.260	0.278
a_{is}^2					0.258	0.212
$b_{i,s-1}$					0.318	0.350
$b_{i,s-1}^2$					0.209	0.185
$a_{i,s-1}$					0.278	0.278
$a_{i,s-1}^2$					0.219	0.224
$b_{i,s-2}$					0.288	0.318
$b_{i,s-2}^2$					0.228	0.220
$a_{i,s-2}$					0.270	0.236
$a_{i,s-2}^2$					0.206	0.206
$b_{i,s-3}$					0.254	0.285
$b_{i,s-3}^2$					0.279	0.228
$a_{i,s-3}$					0.328	0.302
$a_{i,s-3}^2$					0.234	0.188
$b_{i,s-4}$					0.231	0.259
$b_{i,s-4}^2$					0.259	0.215
$a_{i,s-4}$					0.308	0.314
$a_{i,s-4}^2$					0.262	0.231
$b_{i,s-5}$					0.237	0.293
$b_{i,s-5}^2$					0.273	0.213
$a_{i,s-5}$					0.380	0.317
$a_{i,s-5}^2$					0.260	0.220
Total	0.338	0.345	0.492	0.433	0.291	0.268

4.2 Passenger load and alighting

Figure 6 shows predicted against observed loads for the test data set and a specific target station (Västerbroplan). Loads are predicted for all three levels of data availability considered in Section 3.1. The dashed horizontal and vertical lines indicate the seated capacity. The top diagram shows the results using only historical predictors, the middle diagrams add AVL predictors and the bottom diagrams further add APC predictors. In the middle and bottom rows, the left side shows results with 10-min prediction horizon, i.e., predictions from the most recent stop 10 min before departing from the target stop. Due to travel time variations, the source stop may vary between bus trips. The right side shows results with 1-min prediction horizon.

Prediction performance increases with real-time bus data. Historical data tend to underestimate the load on crowded runs, but overestimate the load on the least crowded runs. Thus, historical patterns alone cannot explain well why some buses are highly crowded while others are almost empty. Real-time AVL data improve the bias but show high variance, and performance does not improve much from 10-min to 1-min horizon. Real-time APC data substantially improve the prediction performance, particularly with 1-min horizon.

Figure 7 shows the mean absolute errors (MAE) and mean errors (ME, calculated as in Eq. (34)–(35) on the test data set) of predicted loads across all stops with 10-min and 1-min prediction horizon. Compared with Fig. 5, top left, MAE follow the passenger load magnitude across stops. ME values fluctuate around zero and show no apparent signs of bias. With 10-min horizon, real-time AVL and APC data do not bring benefits before stop 7, since the bus has not yet departed from the first stop. For subsequent stops, real-time AVL predictors bring a substantial reduction in MAE, and real-time APC predictors improve performance somewhat further. With 1-min horizon, real-time APC predictors drastically increase prediction performance compared to the other levels of data availability.

Figure 8 shows predicted against observed alighting counts according to the same principles as in Fig. 6. Due to smaller magnitudes the discrete nature of the numbers is more apparent than for the loads. The contour plots indicate the frequency with which different values occur. The impacts of data availability and prediction horizon follow similar trends as for the loads. Overall, however, prediction performance is lower for alighting counts than for loads.

Figure 9 shows the same metrics as Fig. 7 for predicted alighting counts. Compared to Fig. 7, real-time AVL and APC data are less effective in in-

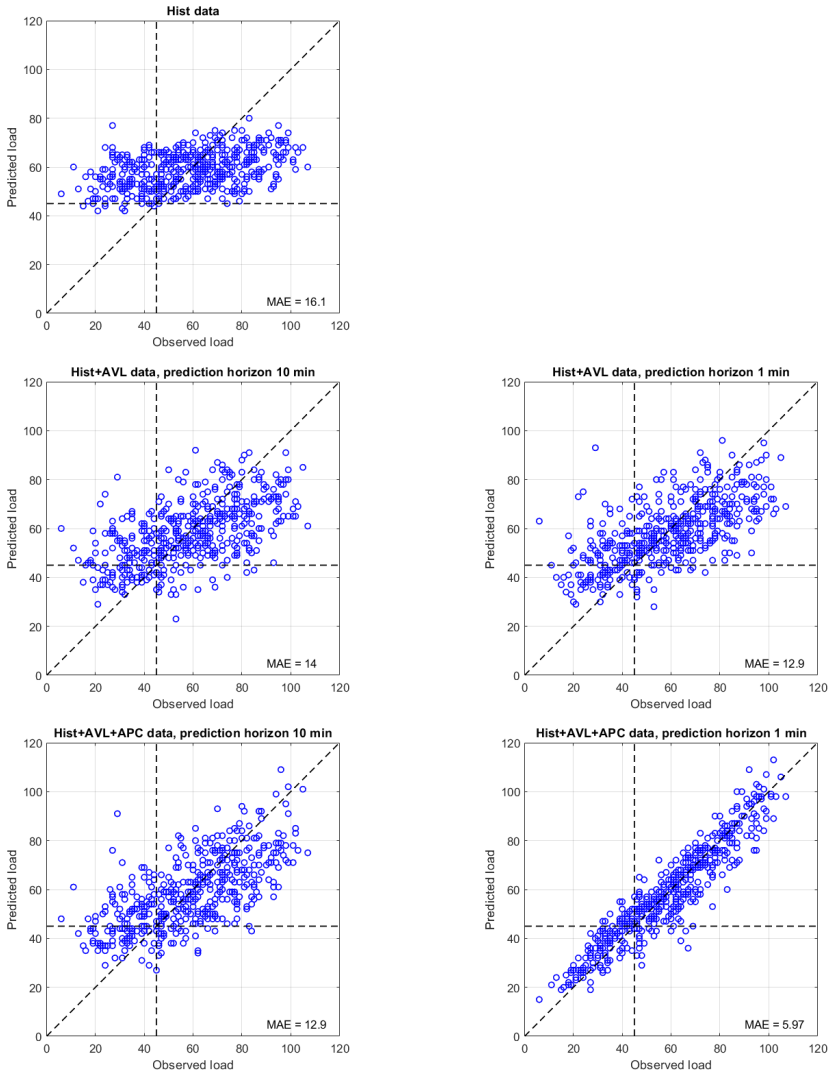


Figure 6: Predicted vs. observed load for test data set, target stop 18 (Västerbroplan). Top: Historical data. Middle: Historical data and real-time AVL data, 10 min. (left) and 1 min. (right) prediction horizon. Bottom: Historical data, real-time AVL and APC data, 10 min. (left) and 1 min. (right) prediction horizon. Horizontal and vertical dashed lines indicate bus seated capacity.

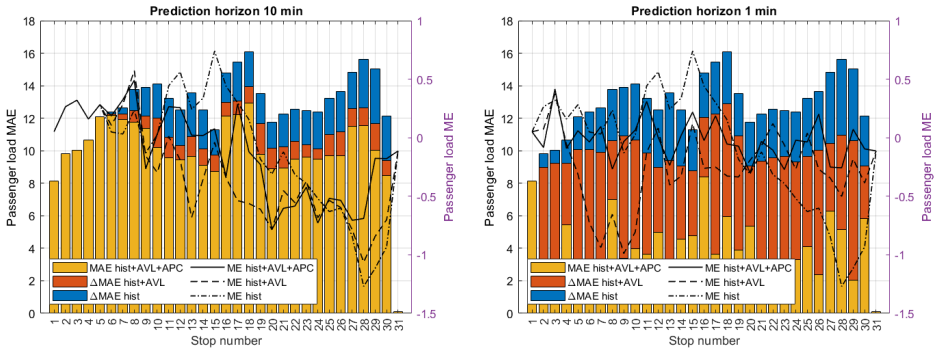


Figure 7: MAE and ME of predicted passenger load across stops. Prediction horizon 10 min (left) and 1 min (right). The top two stacked bars show incremental performance metrics compared to the level below.

creasing prediction performance.

4.3 Probability of seat on boarding

Figure 10 shows the accuracy and ME of the predicted probability to get a seat on boarding across all origin stops. Accuracy is evaluated as the share of observations for which the prediction correctly classifies the crowding with respect to three categories: no available seat certain (PSB $= 0$), seat availability uncertain (PSB $\in (0, 1)$), and available seat certain (PSB $= 1$).

Prediction accuracy is high for the first few stops even with only historical data, since the load rarely reaches the seated capacity. Compared with Fig. 5, top right, accuracy tends to drop at stations where the seat probability is lower. Real-time AVL and APC data increase accuracy, in particular with 1-min prediction horizon where accuracy is around 90% for most stations. With 10-min horizon the ME tends to be positive which suggests that the predictions are biased towards being too optimistic. This is an effect of the non-linear relation with loads and alighting counts. With 1-min horizon there is no evidence of bias.

4.4 Expected travel time standing

Figure 11 shows the MAE and ME of the predicted travel time standing across all origin stations for a trip to the final stop (Gullmarsplan). The top row shows the direct predictions according to Eq. (14) and the bottom row shows the predictions after bias correction according to Eq. (32).

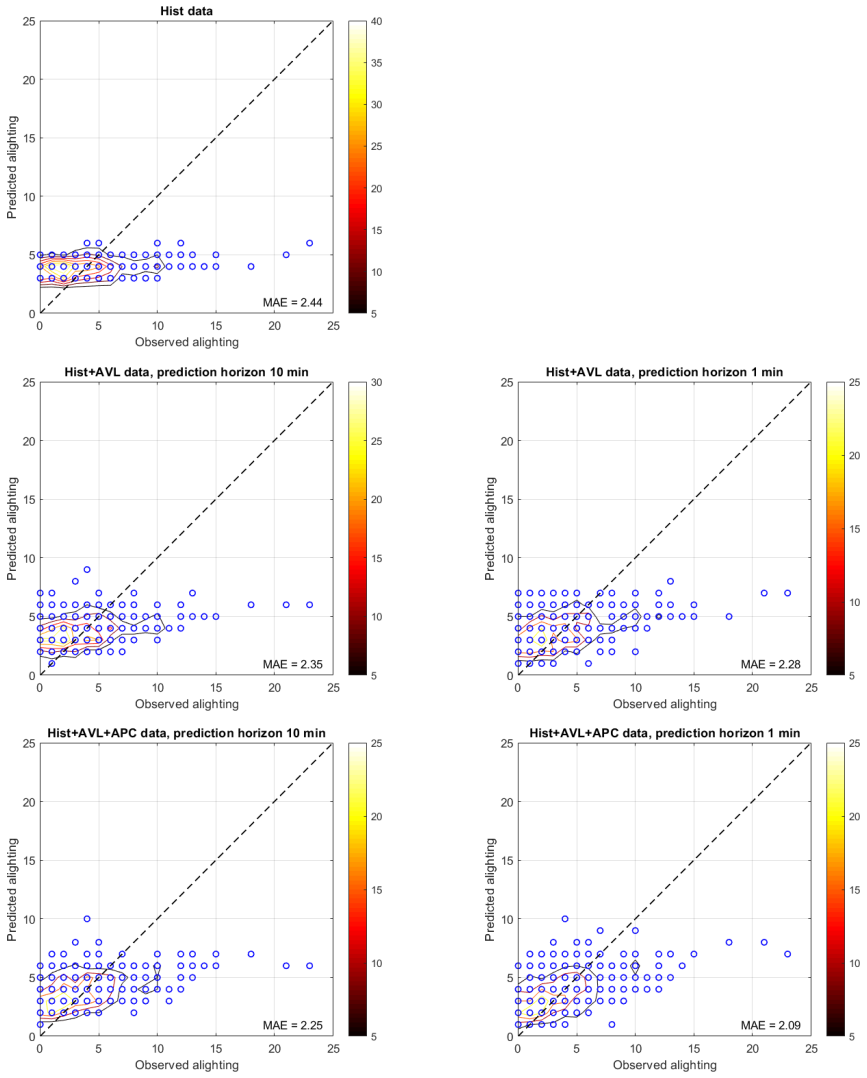


Figure 8: Predicted vs. observed alighting counts for test data set, target stop 18 (Västerbroplan). Top: Historical data. Middle: Historical data and real-time AVL data, 10 min. (left) and 1 min. (right) prediction horizon. Bottom: Historical data, real-time AVL and APC data, 10 min. (left) and 1 min. (right) prediction horizon.

Compared to Fig. 5, bottom left, prediction performance is lower for

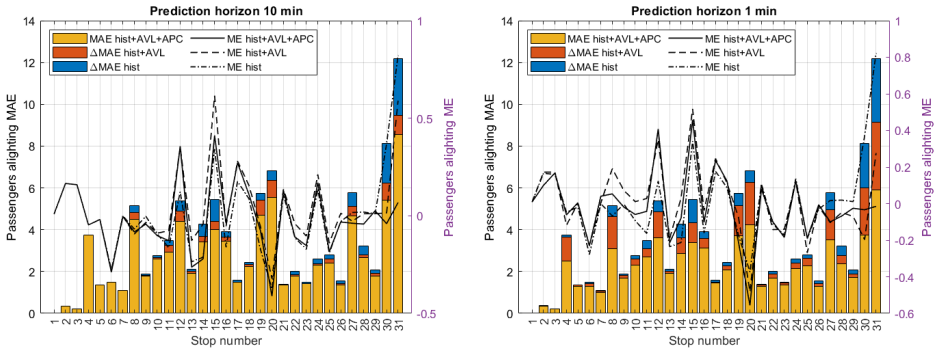


Figure 9: MAE and ME of predicted alighting counts across stops for test data set. Prediction horizon 10 min (left) and 1 min (right). The top two stacked bars show incremental performance metrics compared to the level below.

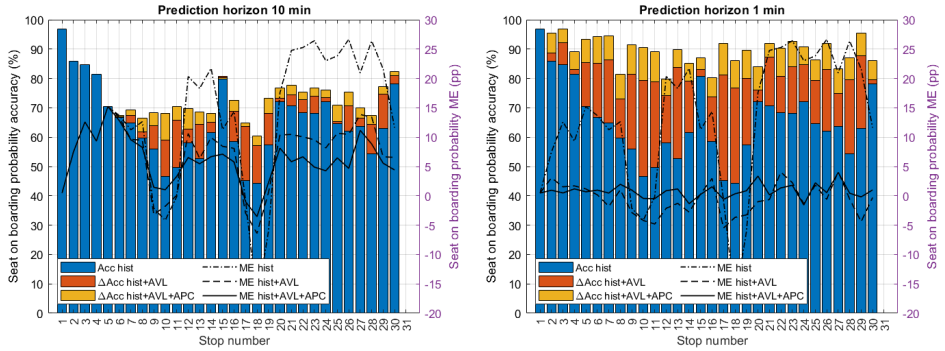


Figure 10: Accuracy (%) and ME (percentage points) of predicted probability of seat on boarding across origin stops. Prediction horizon 10 min (left) and 1 min (right). The top two stacked bars show incremental performance metrics compared to the level below.

origin stops with higher chances of long travel times standing. The reduction in MAE with real-time AVL and APC data is moderate for 10-min horizon, but substantial for 1-min horizon. Without bias correction, predictions tend to be overly optimistic particularly with 10-min horizon. The bias correction manages to reduce ME to practically zero, at the cost of a slight increase in MAE.

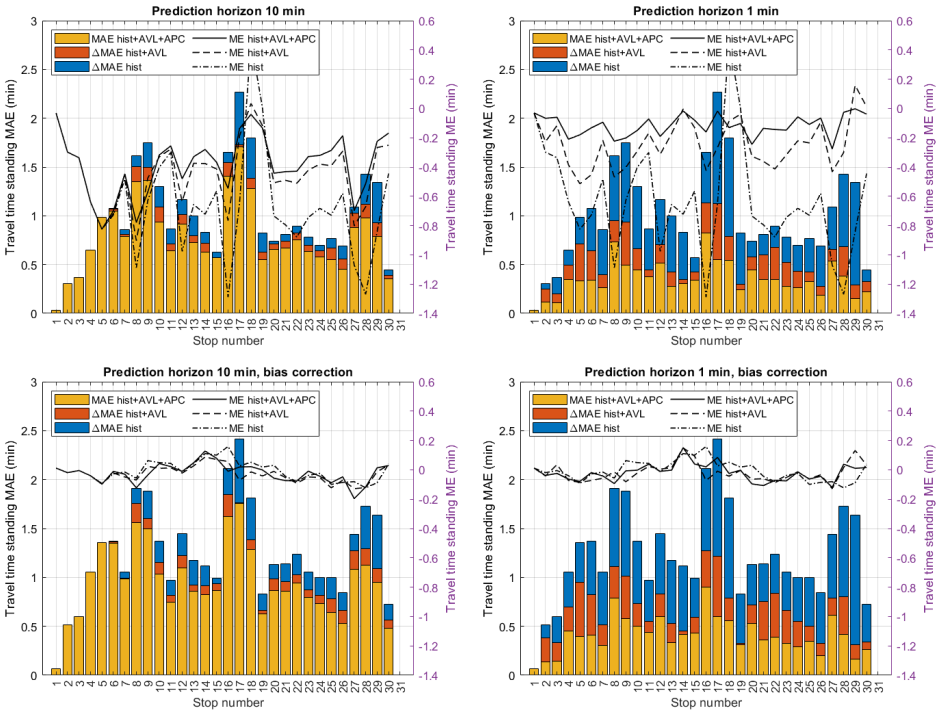


Figure 11: MAE and ME of predicted travel time standing across origin stops for trip to end stop Gullmarsplan. Prediction horizon 10 min (left) and 1 min (right). With (top) and without (bottom) bias correction. The top two stacked bars show incremental performance metrics compared to the level below.

4.5 Excess perceived travel time

Figure 12 shows the MAE and ME of the predicted excess perceived travel time across all origin stations for a trip to the final stop. MAE values are highest for early origin stops due to larger possible variability in trip travel times, and gradually decrease for shorter trips. Compared to the travel time standing, real-time AVL and APC data provide fairly moderate improvements in prediction performance. This is due to the fact that prediction errors grow with the distance between the source and the target station, and the excess perceived travel time is sensitive to crowding conditions during the entire trip.

Without bias correction, predictions tend to be overly optimistic for both prediction horizons. The bias is fairly proportional to the length of the trip in

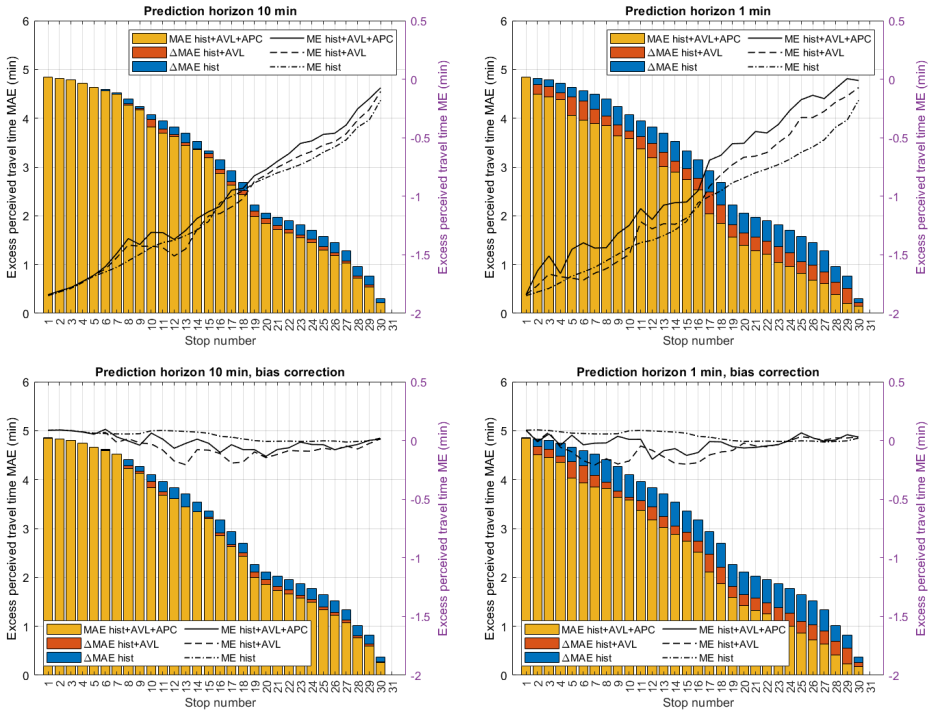


Figure 12: MAE and ME of predicted excess perceived travel time across origin stops for trip to end stop Gullmarsplan. Prediction horizon 10 min (left) and 1 min (right). With (top) and without (bottom) bias correction. The top two stacked bars show incremental performance metrics compared to the level below.

terms of the number of stops. The bias correction reduces ME to practically zero with virtually no effect on the MAE.

4.6 Summary and comparison between lasso and PLS

Table 4 summarizes the prediction performance across all origin stops for the three crowding metrics, the three data availability scenarios, the two prediction horizons and the two prediction methods. As reference, the first row shows the mean observed value of each crowding metric. The summary shows that real-time data improve performance substantially for the probability of seat on boarding and expected travel time standing crowding metrics, while the effect is smaller for the excess perceived travel time. Real-time AVL and APC data combined provide the the highest performance,

Table 4: Mean crowding metric values and prediction performance (accuracy and MAE, respectively) across all origin stops for trip to end stop Gullmarsplan. Probability of seat on boarding (PSB), expected travel time standing (ETTS) and excess perceived travel time (EPTT).

Mean observed crowding	PSB (%) 71.02		ETTS (min) 1.1110		EPTT (min) 5.2686	
	Lasso	PLS	Lasso	PLS	Lasso	PLS
Acc./MAE historical APC	63.2	63.0	0.987	0.993	3.02	3.03
Acc./MAE hist+AVL						
10-min. horizon	68.5	66.9	0.858	0.912	2.90	3.04
1-min. horizon	79.3	78.4	0.548	0.580	2.69	2.88
Acc./MAE hist+AVL+APC						
10-min. horizon	71.4	70.7	0.784	0.819	2.85	3.01
1-min. horizon	86.5	86.3	0.355	0.379	2.44	2.61

but AVL data give a substantial improvement even without APC data. The lasso and PLS regression models performs similarly on historical data, but lasso outpeforms PLS in the real-time data scenarios. This may indicate that the benefits of estimating stop-specific models outweigh the benefits of incorporating correlations among stops in a multivariate model.

5 Conclusions

The paper has proposed a methodological framework for providing personalized, predictive in-vehicle crowding information to public transport travellers. Three crowding metrics, the probability of getting a seat on boarding, the expected travel time standing, and the excess perceived travel time, have been examined. Depending on data availability, the prediction method uses a combination of historical passenger counts, real-time vehicle locations and real-time passenger counts to predict the crowding metrics. A large number of variables are considered; the lasso regularization approach selects a subset of predictors which are used in predictions, while the PLS approach projects the predictors to a lower-dimensional latent space.

The methodology is applied to a case study for a bus line in Stockholm, where all vehicles transmit AVL data but only ca. 20% are equipped with passenger counters. Still, because of its flexibility the method provides predictive crowding information for any trip. The main findings can be summarized as follows:

- The case study reveals that crowding conditions vary substantially

across origin and destination stops. The three personalized crowding metrics show distinctly different spatial patterns from each other and from the passenger load.

- Systematic temporal variations captured in historical load data are useful to provide baseline predictions. Historical predictors are frequently selected in the prediction models even when real-time AVL and APC predictors are available. Interaction terms between time-of-day, weekday and month trends are most frequently selected.

- AVL predictors serve partially as complements and partially as proxies for APC predictors. The source stop headway is the most frequently selected AVL predictor for both loads and alighting counts. The source stop passenger load is the most frequently selected APC predictor.

- The approach of predicting personalized crowding metrics from loads and alighting counts creates some optimistic bias, which can be removed with a simple correction term at little or no cost to MAE.

- For the probability of seat on boarding and the expected travel time standing crowding metrics, real-time AVL and APC predictors provide substantial improvements in prediction performance, in particular for the 1-min prediction horizon. The effects are smaller for the excess perceived travel time crowding metric.

The results indicate that personalized, predictive crowding information that is robust to varying data availability can be provided with sufficient accuracy to be useful to travellers. When available, real-time APC predictors always improve prediction performance. Even without real-time APC data, real-time AVL predictors can substantially improve prediction performance for the probability of seat on boarding and expected travel time standing crowding metrics. This suggests that useful real-time crowding information can be provided also in public transport systems where real-time passenger load data are not yet available.

The results are of value for agencies and operators in order to increase the attractiveness and capacity utilization of public transport. Experience with RTCI in bus systems is still limited, but pilot studies [18] and simulation experiments [37, 36] in metro systems suggest that RTCI can be an effective means to reduce in-vehicle crowding and passengers' discomfort, increase service performance and reduce operation costs.

The presented research can be extended in several directions. An area of further work is to evaluate whether more elaborated prediction methods

can increase performance. In particular, methods that can incorporate possible non-linear effects as well as multivariate correlations between loads and alighting counts at adjacent stops in a transparent way are of interest. Predictions under abnormal conditions such as special events, disruptions etc. is a particular area where novel modelling approaches may be required. Prediction accuracy for the probability of getting a seat on boarding may be increased by formulating it as a classification problem, although this would double the predictions needed if the other crowding metrics are also to be computed.

Another direction for extension is to explore the potential of historical and real-time AFC data for personalized crowding prediction. In particular, information about travellers' origins and destinations could improve the alighting count and in-vehicle seat allocation models. An interesting topic for investigation is also how to design interfaces for RTCI to disseminate the information to travellers in an understandable and useful way.

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